

Git Interview Questions

1. What is Git?

Git is a **distributed version control system** used to track changes in source code and enable multiple developers to collaborate efficiently.

2. What is the difference between Git and GitHub?

Git	GitHub
Version control tool	Cloud platform for hosting Git repositories
Works locally	Works on the internet
Tracks code changes	Collaboration platform

3. What is a repository?

A repository (repo) is a storage location that contains project files and the complete history of changes.

4. What is a commit?

A commit is a snapshot of changes made to the files in the repository.

5. What are the stages in Git?

Stage	Description
Working Directory	where files are edited
Staging Area	files prepared for commit
Repository	committed changes stored

6. What is the Git workflow?

git clone
git add
git commit
git push

7. What is the difference between git pull and git fetch?

A branch is an **independent line of development**.

8. What is a merge conflict?

A merge conflict occurs when **two branches modify the same part of a file** and Git cannot automatically merge them.

9. How do you resolve merge conflicts?

git status

Edit the conflicted file.

git add filename

git commit

10. What is git reset?

git reset moves the branch pointer to a previous commit and optionally modifies the staging area and working directory.

11. What is git stash?

git stash temporarily saves uncommitted changes.

12. What is a Pull Request (PR)?

A pull request is a request to merge changes from one branch into another on GitHub.

13. What is a fork?

A fork is a copy of someone else's repository in your GitHub account.

14. What is GitHub Actions?

GitHub Actions is a CI/CD automation tool used to build, test, and deploy code automatically.

15. How do you clone a repository?

git clone <https://github.com/user/repo.git>

16. How do you check commit history?

git log

17. How do you undo the last commit?

git reset --soft HEAD~1

18. How do you revert a pushed commit?

git revert <commit-id>

19. How do you remove a file from Git tracking?

git rm --cached filename

Common commands In Git

git init

git clone

git status

git add

git commit

git push

git pull

git branch

git checkout

git merge